

VendorFlow AI: Autonomous Vendor Onboarding System

VendorFlow AI is an intelligent, automated vendor and subcontractor onboarding pipeline built on the SAP Cloud Application Programming (CAP) Model. It replaces manual compliance reviews and data entry with an **Agentic AI pipeline** and a **Reinforcement Learning (RL) routing engine**.

Project Overview

When a vendor uploads a registration document, Commercial Agreement, or Subcontractor CV, VendorFlow AI automatically:

1. **Scans** the document for security threats.
2. **Extracts** structured data using Google Gemini Multimodal AI.
3. **Validates** the document against a strict 6-point enterprise compliance policy.
4. **Automatically emails** the vendor if required fields or compliance clauses are missing.
5. **Enriches** the data with parallel KYC and Sanctions checks.
6. **Predicts** the correct workflow routing (Auto-Approve, Compliance Review, Finance Review) using a Reinforcement Learning agent.

Architecture & Tech Stack

- **Frontend (Presentation):** Custom React SPA (Vite + Tailwind CSS). It replaces standard SAP Fiori Elements to provide a dynamic, real-time Kanban-style queue dashboard.
- **Backend (Orchestration):** SAP CAP (Node.js). Exposes an OData V4 API with atomic database transactions (`cds.tx`) to prevent race conditions.
- **Database:** SQLite (Local Development) / Ready for SAP HANA Cloud (Production).
- **Agentic AI Layer:** Google Gemini 2.5 Flash via `@google/generative-ai`. Reads raw PDFs/Images natively via multimodal inline data to extract compliance data without relying on flawed text-parsers.
- **Reinforcement Learning (RL) Engine:** Python FastAPI microservice utilizing a Soft Actor-Critic (SAC) network built with PyTorch.

The Agentic AI Pipeline

The AI layer doesn't just extract text; it understands **context**. The prompt dynamically adjusts its required fields based on the document category:

- **Commercial Agreements:** Checks for MSA, SOW, NDA, Data Privacy Clauses, and BCP/DR capabilities.
- **Resource/Subcontractor Documents:** Checks for Detailed CVs, Government IDs, and Background Verification (BV) clearances.
- **Vendor Master Forms:** Extracts Tax IDs, Bank Routing numbers, and Company addresses.

The "Action-Required" Loop: If the AI detects that critical compliance data is missing, the backend halts the pipeline, moves the vendor to an **Action-Required** queue, and automatically dispatches a personalized email to the vendor detailing exactly which documents they need to re-upload.

Reinforcement Learning & Shadow Mode

The RL agent currently operates in **Shadow Mode**. It evaluates a normalized 16-feature **StateVector** (which includes AI confidence scores, Sanctions risks, and KYC validity) and makes a routing prediction (e.g., **Route-Finance** or **Auto-Approve**).

Instead of routing the vendor automatically, it waits for a human reviewer to make the final decision in the React dashboard.

- If the human agrees with the AI, the agent receives a **Reward of +1**.
- If the human disagrees, the agent receives a **Reward of -1**.

A weekend batch-training script (**train.py**) uses these human-labeled reward signals to constantly retrain and improve the AI's routing accuracy over time.

Roadmap to Scalability (Production Readiness)

To take this from a robust PoC to a massively scalable enterprise deployment, the following architectural upgrades are recommended:

1. Asynchronous Messaging (Event Mesh)

Currently, the pipeline runs via Node's **setImmediate**. In production, the **VendorDocument** upload should publish a **CloudEvent** to SAP Event Mesh or Kafka. This ensures that if the AI extraction takes longer than expected (or fails), the message is queued and safely retried without blocking the main OData thread.

2. Database Migration (SAP HANA Cloud)

The current implementation uses SQLite. Deploying to SAP HANA Cloud will provide enterprise-grade persistence, native JSON querying for the `StateVector`, and the ability to handle millions of training batches for the RL agent.

3. Email & Security Integrations

- **Notifications:** Swap the local `nodemailer` SMTP setup for the **SAP Cloud SDK Mail Client**, utilizing BTP Destinations to securely route emails through the corporate Exchange server.
- **Malware Scanning:** Replace the stubbed `_runMalwareScan` function with an integration to the **SAP BTP Malware Scanning Service**.

4. Transitioning RL out of Shadow Mode

Once the RL Agent achieves a >95% human-agreement accuracy rate in Shadow Mode, the SAP CAP orchestrator can be updated to trust the `PredictedAction` directly. The human review step will be bypassed for low-risk vendors (`Auto-Approve`), achieving true Zero-Touch Onboarding.

5. AI usage

Also use SAP AI Generative HUB from SAP BTP as an AI extractor as the documents are very sensitive so it should pass through secured pipelines rather than some public api keys call.